

Applied Linear Algebra

Linear Functions

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We write $f : \mathbb{R}^n \rightarrow \mathbb{R}$ to mean that the function f maps real n -vectors to real numbers.

Function f is a scalar-valued function of n -vectors.

Example

$$f(\mathbf{x} : \mathbb{R}^n) = \mathbf{1}_n^T \mathbf{x}$$

Consider the vector inner product function

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \cdots + a_n x_n$$

where $\mathbf{a}$ is a fixed n -vector, and $\mathbf{x}$ is a variable n -vector.

The function f is called a *linear function* of $\mathbf{x}$.

The vector $\mathbf{a}$ is called the *coefficient vector* of the linear function f .

$\mathbf{a}$ is also called the *weight vector* of the linear function f .

A linear function satisfies the following two properties:

- *Additivity.* $f(\mathbf{x} + \mathbf{y}) = f(\mathbf{x}) + f(\mathbf{y})$ for all n -vectors $\mathbf{x}$ and $\mathbf{y}$.
- *Homogeneity.* $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$ for all n -vectors $\mathbf{x}$ and all scalars α .

The vector inner product function

$$f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} = a_1 x_1 + \cdots + a_n x_n, \quad \mathbf{a} : \mathbb{R}^n, \quad \mathbf{x} : \mathbb{R}^n$$

is linear.

Definition

A function f is *linear* if it satisfies the *superposition principle*:

$$f(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f(\mathbf{x}) + \beta f(\mathbf{y})$$

for all n -vectors $\mathbf{x}$ and $\mathbf{y}$, and all scalars α and β .

Given the inner product function $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$, we can verify that it is linear:

$$\begin{aligned} f(\alpha \mathbf{x} + \beta \mathbf{y}) &= \mathbf{a}^T(\alpha \mathbf{x} + \beta \mathbf{y}) \\ &= \mathbf{a}^T(\alpha \mathbf{x}) + \mathbf{a}^T(\beta \mathbf{y}) \\ &= \alpha(\mathbf{a}^T \mathbf{x}) + \beta(\mathbf{a}^T \mathbf{y}) \\ &= \alpha f(\mathbf{x}) + \beta f(\mathbf{y}) \end{aligned}$$

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Unit 1